

Auto Release Control (ARC) Dynamic Peak Limiting & Gain Smoothing in Time-Dilated Audio Engines

Time Dilation DAW Research Group — Version 0.0.1

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1 Overview

In relativistic time-dilated audio synthesis, dynamic time warping and feedback cascades can produce transient amplitude spikes exceeding standard 0.0 dBFS audio ceilings. To prevent output clipping and digital inter-sample distortion while maintaining acoustic warmth, the `out~` (`OutNode`) master object incorporates an automatic **ARC Safety Peak Limiter** operating at a ceiling of -1.5 dBFS.

2 Ceiling & Gain Threshold Math

The linear ceiling threshold C for a peak ceiling of -1.5 dBFS is defined by:

$$C = 10^{\frac{-1.5}{20}} \approx 0.8413951 \quad (1)$$

For each audio sample $s \in [0, N - 1]$ in a stereo block, the instantaneous peak magnitude $p[s]$ is computed as:

$$p[s] = \max(|s_L[s]|, |s_R[s]|) \quad (2)$$

The target gain multiplier $g_{\text{target}}[s]$ required to constrain $p[s]$ within the ceiling C is:

$$g_{\text{target}}[s] = \begin{cases} 1.0 & \text{if } p[s] \leq C \\ \frac{C}{p[s]} & \text{if } p[s] > C \end{cases} \quad (3)$$

3 Two-Stage Attack & Adaptive ARC Release Filter

To suppress transient clipping instantly without introduced harmonic distortion or audible pumping during sustained signals, the limiter uses an **Auto Release Control (ARC)** mechanism.

3.1 Attack Phase

The attack filter operates at an ultra-fast time constant $\tau_{\text{att}} = 0.05$ ms ($50 \mu\text{s}$):

$$\alpha_{\text{att}} = 1.0 - \exp\left(-\frac{1.0}{f_s \cdot 0.00005}\right) \quad (4)$$

When $g_{\text{target}}[s] < g[s - 1]$, the limiter triggers attack attenuation:

$$g[s] = g[s - 1] + \alpha_{\text{att}} \cdot (g_{\text{target}}[s] - g[s - 1]) \quad (5)$$

3.2 Adaptive ARC Release Phase

When $g_{\text{target}}[s] \geq g[s-1]$, the release time constant dynamically adjusts based on the transient intensity ratio $R = p[s]/C$:

$$\tau_{\text{rel}}(R) = \begin{cases} 10 \text{ ms} & \text{if } R < 1.5 \quad (\text{Fast recovery for transient spikes}) \\ 300 \text{ ms} & \text{if } R \geq 1.5 \quad (\text{Smooth recovery for sustained peaks}) \end{cases} \quad (6)$$

The release smoothing coefficient α_{rel} is computed sample-by-sample:

$$\alpha_{\text{rel}} = 1.0 - \exp\left(-\frac{1.0}{f_s \cdot (\tau_{\text{rel}}/1000.0)}\right) \quad (7)$$

$$g[s] = g[s-1] + \alpha_{\text{rel}} \cdot (g_{\text{target}}[s] - g[s-1]) \quad (8)$$

4 Output Signal Scaling

The final limited stereo sample pair $(y_L[s], y_R[s])$ delivered to the DAW output buffer is:

$$y_L[s] = s_L[s] \cdot g[s], \quad y_R[s] = s_R[s] \cdot g[s] \quad (9)$$

This guarantees zero hard clipping at ± 1.0 , complete protection of downstream audio devices, and transparent ARC release dynamics.